

MSME IDEA HACKATHON 6.0Reference No. :- **26INC06TN033202**

1. Business Incubator Name	Dr. S. Thanga Kasi Rajan and Dr. D. Prince Winston
2. Business Incubator Email ID	hodmech@kamarajengg.edu.in
3. Business Incubator Mobile Number	9942670301
4. HI/BI State	TAMIL NADU
5. HI/BI Name	KAMARAJ COLLEGE OF ENGINEERING AND TECHNOLOGY
6. Theme	Industry 4.0 and 5.0
7. Idea Sector	Miscellaneous Sector (Environment, Forests, Water & Sanitation; Foods, Beverages, FMCG, Consumer Goods; Infrastructure, Construction, Housing; IT, ITES, Electronics, White Goods, Telecommunication; Metals, Engineering, Machinery, Automation and Transportation, Automotive, E Vehicles, Railways, Aviation, UAV and any other sub-sector)
8. Title of proposed idea/innovation	AI-Based Quality Inspection System Using Camera Vision
9. Briefly explain newness/uniqueness of the innovation	While an industrial camera-based machine learning solution can cost thousands, this one is affordable and modular, employing relatively inexpensive cameras and open-source machine learning frameworks, which allows small and midsize manufacturers to deploy automated inspections without needing costly solutions. This technology distinguishes itself by bringing all three elements together into a closed loop – defect detection, automatic quality score and rejection, unlike any other available solution, which focuses only on defect detection. Another important feature of the AI model is that it is retrainable thus, MSMEs can use it without being experts in the field of AI and without having a particular product category in mind, whether it is textile, plastic or food packaging.
10. Concept & Objective	• Automated AI-based inspection: A camera captures product images, and an AI model detects defects such as scratches, cracks, and surface irregularities in real time. • Real-time quality assessment: The system generates a quality score for each product and automatically rejects defective items using an actuator. • Improved efficiency for MSMEs: Reduces manual inspection, human error, and inspection time while ensuring consistent, accurate, and scalable quality control.
11. Specify the potential areas of application in industry/market in brief	The system offers great potential for application in the textile industry for the purpose of identifying problems related to weaving of fabric, alignment of printing and stains on fabric, and in the plastic industry for identifying mold defects, surface cracks and dimensioning problems in molded or extruded items. Within the food packaging sector, it could help ensure that seals are intact, labels are in the right place, and that packages have not been damaged or contaminated prior to shipment. In addition to the main industries, the solution can also be applied to other industries including electronics assembly, pharmaceuticals, and FMCG packaging lines.

12. Briefly provide the market potential of idea/innovation	There is a good number of MSMEs in India in the textile, plastics, and food processing sectors who use manual inspection methods to perform quality checks. In light of the ever-increasing demands of strict quality standards from local and international consumers, there is a need for companies to upgrade their operations without having to make huge investments in expensive industrial vision systems. There exists a high demand for an AI-based inspection system that is affordable and easy to implement and which can either be sold as a one-off product or an inspection as a service offering.
13. Uploaded Block diagram/ flow chart/ Circuit Diagram/Pictures	View/Download